Mapping Forest Canopy Height at Large Scales Using ICESat-2 and Landsat: An Ecological Zoning Random Forest Approach

Zhaocong Wu and Fanglin Shi

***Abstract*— Forest canopy height (FCH) is a crucial indicator in the calculation of forest biomass and carbon sinks. There are various methods to measure FCH, such as space-borne light detection and ranging (LiDAR), but their data are spatially discrete and do not provide continuous FCH maps. Therefore, an FCH estimation method that associates sparse LiDAR data with spatially continuous variables is required. The traditional approach of constructing a single model overlooks the spatial variability in forest growth, which will limit the FCH accuracy. Considering the distinct nature of forest in different ecological zones, the following hold. First, we proposed an ecological zoning random forest (EZRF) model in 33 ecological zones in China. Compared with the total zone RF (TZRF) model, the EZRF model showed a greater potential, which was 21.5%–36.5% more accurate than the TZRF model. Second, we analyzed a total of 62 variables related to forest growth, including Landsat variables and ancillary variables (forest canopy cover, bioclimatic, topographic, and hillshade factors). An insight into variable selection in FCH modeling was provided by analyzing the prediction accuracy of FCH under different categorical variables and analyzing the importance of variables in different ecological zones. Third, finally, we produced a 30-m continuous FCH map by the EZRF model. Compared with the airborne LiDAR data, the FCH prediction results produced a root mean square error (RMSE) of 2.50–5.35 m, which were 41%–72% more precisely than the Global FCH, 2019. The results demonstrate the effectiveness of our proposed method and contribute to the study of forest carbon sinks.**

***Index Terms*— Cloud, ecological zonal modeling approach, forest canopy height (FCH), hillshade variables, Ice, and Land Elevation Satellite-2 (ICESat-2) ATL08, Landsat variables.**

I. Introduction

FOREST canopy height (FCH) is an important attribute in forest resource surveys, forest productivity, and biodiversity [1], and a key factor in estimating forest biomass and forest carbon sinks [2], [3]. Consistent, accurate, and large-scale FCH estimates are essential for estimating forest-related carbon emissions, analyzing forest degradation, and quantifying the effectiveness of forest restoration programs.

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The authors are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: zcwoo@whu.edu.cn; shifanglin@whu.edu.cn).

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In forestry, FCH is one of the most prominent tree measurement factors, that is, the distance from the top of the canopy to the ground [4]. Traditionally, tree heights have been measured in the field, but this method is labor intensive. Lately, airborne light detection and ranging (LiDAR) has become an accurate way to measure spatially continuous FCH [5], [6]. However, it is also a challenge in FCH mapping on a large scale, due to its expensive data collection, and differences in data collection time and parameters on large scales [7]. Spaceborne LiDARs can quickly yield consistent FCH data and are more suitable than airborne LiDARs for FCH mapping over large areas. Currently, there are four spaceborne LiDAR systems in the world that provide scientific data: the Ice, Cloud, and Land Elevation Satellite (ICESat) [8], launched by NASA in 2003; ICESat-2 [9] and Global Ecosystem Dynamics Investigation (GEDI) [10], launched in 2018; and Gao Fen 7 (GF-7), launched by China in 2019 [11]. Among them, the ICESat, GEDI, and GF-7 laser altimeter systems are linear systems with low repetition frequency, and ICESat-2 is the only satellite in the world that carries the Advanced Topographic Laser Altimeter System (ATLAS), a photon-counting LiDAR system with high repetition frequency, which greatly expands the 3-D information detection capability [9]. ICESat-2 has notable advantages compared with ICESat, and the laser footprints of ICESat-2 ATLAS are ≈0.7-m apart and 17 m in diameter along the direction of the orbit, enabling continuous detection along the direction of the orbit; it cannot provide wall-to-wall FCH directly due to its profiling configuration and somewhat limited detection capabilities. Therefore, a method of associating the sparse LiDAR footprint with spatially continuous variables is needed.

The use of spatially continuous satellite imagery to upgrade satellite LiDAR FCHs from discrete point states to region-continuous levels has become a hot topic in the large-scale FCH mapping. Various mathematical methods are currently available for upgrading FCHs with additional attribute variables [12], [13], [14], [15], [16], [17]. They mostly estimate the FCH by using a single model over a large area. However, due to spatial variability in forest growth [18], the traditional construction of a single model will limit the FCH estimation accuracy. Also, a single FCH model applied to a large coverage area may not be appropriate for the full range of sites in all biomes. Therefore, developing trained and highly modeled patches, segmenting or classifying larger

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learning regions based on common ecological parameters has become a common approach. Lefsky [19] divided moderate resolution imaging spectroradiometer (MODIS) images into polygons for forest height modeling according to spectrum and texture, with an average size of 25 km^2. Potapov et al. [20] segmented Landsat Global Land Analysis and Discovery team (GLAD) Analysis Ready Data (ARD) [21] by $1° \times 1°$ tiles in forest height modeling. However, too fine-grained patches increase the computational burden of modeling forest heights and are not suitable for practical applications in forestry. Wang et al. [22] took North America, South America, Europe, Africa, Asia, Oceania, and Siberia as the division regions for forest height model training. However, too coarse a partition often ignores local climatic factors and forest types, affecting the accuracy of FCH predictions. Large-scale differences in climatic factors and forest growth conditions may lead to different modeling relationships between FCH data and ancillary variables. Considering the distinct nature of forest growth in different ecological zones, building ecological zone-based models is one of the approaches to improve the accuracy of FCH estimation.

Optical remote sensing data can provide distinct spectral information of the forest canopy, which is commonly used in FCH modeling. For example, Landsat multispectral images, from which spectral information, vegetation indices, spatial textures, and other variables related to the forest canopy can be extracted, were commonly used as the FCH predictor variables [23], [24], [25]. Li et al. [25] combined the current ICESat-2 ATL08 data with Landsat variables (including six spectral bands, three vegetation indices, and three texture variables) generated a 30-m continues FCH map of the Inner Mongolian region of China and indicated these spectral variables can positively contribute to the FCH prediction. Except for the optical remote sensing variables, biophysical variables [17], [22], such as climate and topography [26], [27], that affects forest growth have been used in FCH estimation tasks. For example, Simard et al. [16] estimated the global FCH map with spatially denser ancillary variables (forest type, tree cover, elevation, and climatology maps). Their FCH maps showed a higher accuracy when compared with the FCH maps [19] estimated from optical remote sensing variables alone. Due to the presence of spectrally similar features in different tree species, extremely mixed complex forest structures, and discontinuous spatial forest distribution in different ecological zones [28], selecting a single type of variable to predict FCH over a large area can be challenging. Moreover, it is necessary to scientifically infer which variables are essential for modeling FCH, and whether modeling multivariable participation is one way to improve the accuracy of FCH estimation. Therefore, it is necessary to measure the variable importance of forest height estimation by considering a variety of variables related to forest growth in a comprehensive way and explore the performance of tree height estimation under different combinations of variable sets.

Based on the above, we proposed an FCH modeling method, which we designated the ecological zoning random forest (EZRF) model. The eastern monsoon ecozone of China, which comprises 33 different ecological zones, was chosen as the experimental area for the EZRF. In order to demonstrate the effectiveness of the EZRF method for large region FCH estimation, we also conducted comparative experiments, that is, building a total zone RF (TZRF) model. The main work carried out are as follows: 1) ICESat-2 ATL08 data were processed to obtain high-precision FCH samples; 2) conducted a comparison analysis of the FCH estimation method by building an EZRF model and a TZRF model for the eastern monsoon ecozone of China, explored the most accurate method, and produced the FCH map; 3) the important predictor variables in forest height modeling were analysis, mainly focus on the driving factors in the modeling of FCH and the internal mechanism of predictive variables in different ecological zones, and make the model more explanatory; and 4) the accuracy of the FCH map was evaluated using airborne LiDAR data and the Global FCH, 2019 [20].

II. Materials

A. Study Area

We focused our experiments on the eastern monsoon ecozone of China, which is outlined in Fig. 1(a) (red) and covers an area of 2.5 million km^2. The eastern monsoon ecozone borders both the Tibetan plateau and the ocean, with high temperatures and rainy conditions in summer, cold, and dry weather in winter. From north to south, the ecozone spans the semiarid mid-temperate monsoon climate zone, the semihumid warm temperate monsoon climate zone, and the subtropical monsoon climate zone, with distinct thermal and hydrothermal gradients. Vegetation coverage varies significantly along the climate gradient. Many types of vegetation are found in the region, which accounts for 84.9% of China's total forest area. According to the global land cover product with a fine classification system at 30-m resolution in 2020 (GLC_FCS30-2020 [29]), the forest in this study was defined as tree-cover percentage greater than 15% and tree height greater than 3 m. Forest types include evergreen broad-leaved forest (EBLF), deciduous broad-leaved forest (DBLF), evergreen needle-leaved forest (ENLF), deciduous NLF (DNLF), and mixed forest (MF). We selected BLF (BLF: EBLF + DBLF), NLF (NLF: ENLF + DNLF), and MF that were selected as forest types for FCH mapping.

B. Airborne LiDAR Data

The airborne LiDAR data were collected in five regions within the study site, as shown in Fig. 1(c)–(g). The first block of data was for Libo, located in Guizhou Province [Fig. 1(c)], which is a karst evergreen BLF. The second two blocks of data were for Yongzhou in Hunan Province, the Mountainous Evergreen BLF [Fig. 1(d) and (e)]. The last two blocks of data were for Yichang in Hubei Province and were mainly covered by mountainous and hilly evergreen BLFs [Fig. 1(f) and (g)]. The LiDAR system used was an SZT-250 mounted on an airborne DJI M600. The airborne LiDAR data were acquired in July and August of 2021 with an average point density of 40 points/m^2, and the average ground speed was 28.8 km/h relative to the flight altitude

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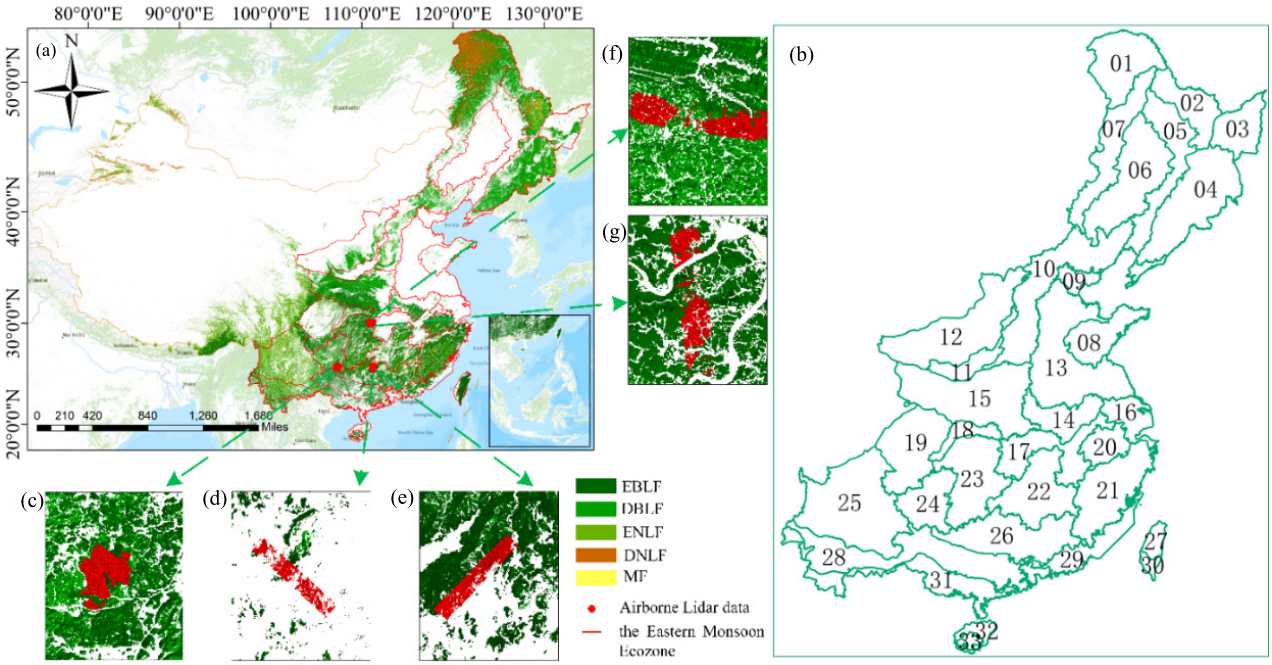


Fig. 1. General overview of the study area. (a) Distribution of forest types and airborne LiDAR data (c)–(g). (b) Location of the eastern monsoon ecozone and the 33 ecological zone distribution.

TABLE I
BASIC INFORMATION OF THE AIRBORNE LiDAR DATA

Index	Study site	Ecological zone	Date	Point Density	Flight Height	Total trees
(a)	Libo, Guizhou	Karst Evergreen BLF	Jul-Aug 2021	40 pts/m2	50.2 m	59555
(b)	Yongzhou, Hunan	Mountainous Evergreen BLF	Jul-Aug 2021	40 pts/m2	50.2 m	17970
(c)	Yongzhou, Hunan	Mountainous Evergreen BLF	Jul-Aug 2021	40 pts/m2	50.2 m	48839
(d)	Yichang, Hubei	Mountainous and Hilly Evergreen BLF	Jul-Aug 2021	40 pts/m2	50.2 m	58906
(e)	Yichang, Hubei	Mountainous and Hilly Evergreen BLF	Jul-Aug 2021	40 pts/m2	50.2 m	37270

of 50.2 m (Table I). The data were used in the Standard LiDAR storage format (.las), and all data were processed at the ENVI LiDAR data processing and analysis module. The airborne LiDAR data projection coordinates were georeferenced to the Landsat Operational Land Imager OLI data. The total number of trees in each block are shown in Table I. Finally, we produced the validation data by averaging the tree height according to the size of 30 × 30 m.

C. ATL08 Data

NASA provides 14 ATLAS data products, including products for ice, atmosphere, vegetation, land, and ocean. All the ATLAS products are available from the National Snow and Ice Data Centre (NSIDC; https://nsidc.org/data/icesat-2/tools). ATL08 (Land and Vegetation Forest Height Product) is one of the ICESat-2 ATLAS products, providing an estimate of terrain and canopy height on a spatial scale with a 100-m segment size along the trajectory direction [25]. ATL08 (Level 3A) was used in our study from 2018 to 2020, and the vegetation height product (h_canopy), which represents the 98% height of all the individual relative canopy heights [30], was used as a dependent variable in the FCH model. When using the ICESat-2 ATL08 data, a series of filtering processes were applied for data accuracy.

1) ATL08 data from nonforested areas were excluded. As our research object was forests, we only selected ATL08 data corresponding to BLF, NLF, and MF land cover types.

2) Data from ATL08 with insufficient ground and canopy photons were removed. According to the theoretical basis of the ATL08 algorithm, if there are fewer than 50 photons on the ATL08 segment, the topographic height does not accurately represent the topographic surface. If there are more than 50 photons on the segment but there are not enough terrain photons, it is impossible to determine the terrain height (due to the dense forest canopy) [31]. Therefore, to avoid the problem of terrain height estimation error due to insufficient ground photons, the following criteria were chosen: the fraction of both canopy and terrain photons was greater than 5%, and the number of canopy photons was greater than 50.

3) ATL08 data with geolocation errors were excluded to match the 30 m pixel center. Since there were five 20-m ATL03 geosegments at each ATL08 h_canopy 100-m segment [30], longitude and latitude were interpolated to the midpoint of the 100-m ATL08 segment if there were at least three 20-m ATL03 geosegments present in ATL08 segment center with enough canopy photons. In order to preserve the ATL08 data with

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the latitude and longitude of the segment center, we excluded the ATL08 data with no canopy photons on either side of the segment center based on the subset_can_flag parameter and determined the ATL08 data with enough photons present in the center of the 20-m segment.

4) ATL08 data with large terrain height errors and forest height errors were excluded. Since ATL08 h_canopy is the difference in height between the top of the canopy and the terrain surface, the estimation error of the terrain height affects its accuracy. Since the most accurate global terrain height product, SRTM, has an altitude accuracy between 5 and 10 m [29], we selected the lower limit of 5 m as our terrain height uncertainty criterion.

After filtering, we obtained 6.5 million ATL08 h_canopy data in the eastern monsoon zone of China, which were used as response data for the FCH model. We randomly divided the ATL08 h_canopy dataset into three parts for modeling and validation as follows: a training dataset (70%), a model validation dataset (20%), and a forest height map validation dataset (10%).

D. Landsat 8 OLI Data

Landsat satellites have been providing unprecedented spatial coverage since 1972. The data are continuous and are provided free of charge to facilitate scientific research and knowledge advancement [32]. In this study, we collected Landsat 8 OLI Tier 1 surface reflectance multispectral images for the 2017–2019 growing season (June–September) from the Google Earth Engine (GEE; data ID: Landsat/LC08/C01/T1_SR) for FCH prediction, and the data resolution was 30 m. The time series of the Landsat images were synthesized into one median image for the leaf-on season (according to ecological zone), and the clouds in the images were removed using Landsat pixel quality attributes. We used Landsat images to create a set of spectral variables that serve as predictors for the FCH model. Here, we chose the spectral reflectivity in bands B2–B7 (see Table II). The spectral indices included the normalized difference vegetation index (NDVI) [33], differential vegetation index (DVI) [34], ratio vegetation index (RVI) [35], soil-adjusted vegetation index (SAVI) [36], enhanced vegetation index (EVI) [37], and leaf area index (LAI) [38]. The tasseled cap (TC) transformation yielded a set of orthogonal components: brightness (TCB), greenness (TCG), and wetness (TCW) [39]; TC components have been widely used to characterize vegetation conditions [40]. Furthermore, to incorporate spatial variables, we calculated the texture measurements of each band (B2–B7) within the 3×3 pixel kernel [41], [42]. All the variables extracted from the Landsat images are shown in Table II. Landsat variables were all conducted using cloud computing technology on the GEE platform (https://code.earthengine.google.com/), which facilitates parallel computing and big data processing over large study areas.

TABLE II
LANDSAT VARIABLES USED FOR FCH MODELING

Type	Variable	Description
spectral reflectance	B2	Blue reflectance (0.450-0.515 μm)
	B3	Green reflectance (0.525-0.600 μm)
	B4	Red reflectance (0.630-0.680 μm)
	B5	Near infrared reflectance(NIR, 0.845-0.885 μm)
	B6	Shortwave infrared 1 reflectance (SWIR1, 1.56-1.66 μm)
	B7	Shortwave infrared 2 reflectance (SWIR2, 2.1-2.3 μm)
Spectral indices	NDVI	NDVI = (B5 - B4) / (B5 + B4)
	DVI	DVI = B5 - B4
	RVI	RVI = B5 / B4
	SAVI	$\mathrm{SAVI} = 1.5 * [(\mathrm{B5} - \mathrm{B4})/(\mathrm{B5} + \mathrm{B4} + 0.5)]$ $\mathrm{SAVI} = [(\mathrm{B5} - \mathrm{B4})/(\mathrm{B5} + \mathrm{B4} + 0.5)]$
	EVI	$\mathrm{EVI} = 2.5 * (\mathrm{B5} - \mathrm{B4})/(\mathrm{B5} + 6 * \mathrm{B4} - 7.5 * \mathrm{B2} + 1)$
	LAI	$\mathrm{LAI} = (2.5 * (\mathrm{B5} - \mathrm{B4}/\mathrm{B5} + 6 * \mathrm{B4} - 7.5 * \mathrm{B2} + 1)) * 3.618 - 0.118$
	TCB	$\mathrm{TCB} = 0.303 * \mathrm{B2} + 0.279 * \mathrm{B3} + 0.473 * \mathrm{B4} + 0.560 * \mathrm{B5} + 0.508 * \mathrm{B6} + 0.187 * \mathrm{B7}$
	TCW	$\mathrm{TCW} = -0.294 * \mathrm{B2} - 0.243 * \mathrm{B3} - 0.542 * \mathrm{B4} + 0.728 * \mathrm{B5} + 0.071 * \mathrm{B6} - 0.161 * \mathrm{B7}$
	TCG	$\mathrm{TCG} = 0.151 * \mathrm{B2} + 0.197 * \mathrm{B3} + 0.328 * \mathrm{B4} + 0.341 * \mathrm{B5} - 0.712 * \mathrm{B6} - 0.456 * \mathrm{B7}$
Texture	Contrast	Measures the local contrast of an image
	Dvar	Difference variance
	Inertia	Inertia

E. Ancillary Data

1) Climate Data: Climate data are widely used to study the distribution patterns of terrestrial vegetation and can be closely related to FCH [43]. Temperature and precipitation are the two main factors controlling vegetation growth [28]. WorldClim is a global climate database [44] and can be obtained from https://www.worldclim.org/data/worldclim21.html. For this study, WorldClim version 2.1 climate data for 1970–2000 were used. This version was released in January 2020. Bioclimatic data (bio1–bio19) from WorldClim were selected as climate variables, and their spatial resolution was ≈1 km (see [44] for more variable details).

2) Topographic Data: The growth of forest vegetation is closely related to topographical factors. In our study, Shuttle Radar Topography Mission (SRTM) data were used to obtain topographic information related to vegetation growth [45]. We obtained SRTM digital elevation data for version 3 from GEE. Its ID in the GEE platform is "USGS/SRTMGL1_003," with a spatial resolution of 30 m. From the SRTM data, we obtained the elevation data directly and calculated the slope and aspect. Besides, we used six hillshade variables in FCH estimation for the first time, and they were simulated at the azimuths of 0° (hd1), 60° (hd2), 120° (hd3), 180° (hd4), 240° (hd5), and 300° (hd6) by the following equation:

$$\mathrm{Hillshade} = \cos(\mathrm{azimuth} - \mathrm{aspect}) \times \sin(\mathrm{slope}) \times \sin(\mathrm{zenith}) + \cos(\mathrm{zenith}_{\mathrm{rad}}) \times \cos(\mathrm{slope}). \quad (1)$$

To calculate the hillshade variables, five types of input data are required: 1) the raster data of the terrain height; 2) the

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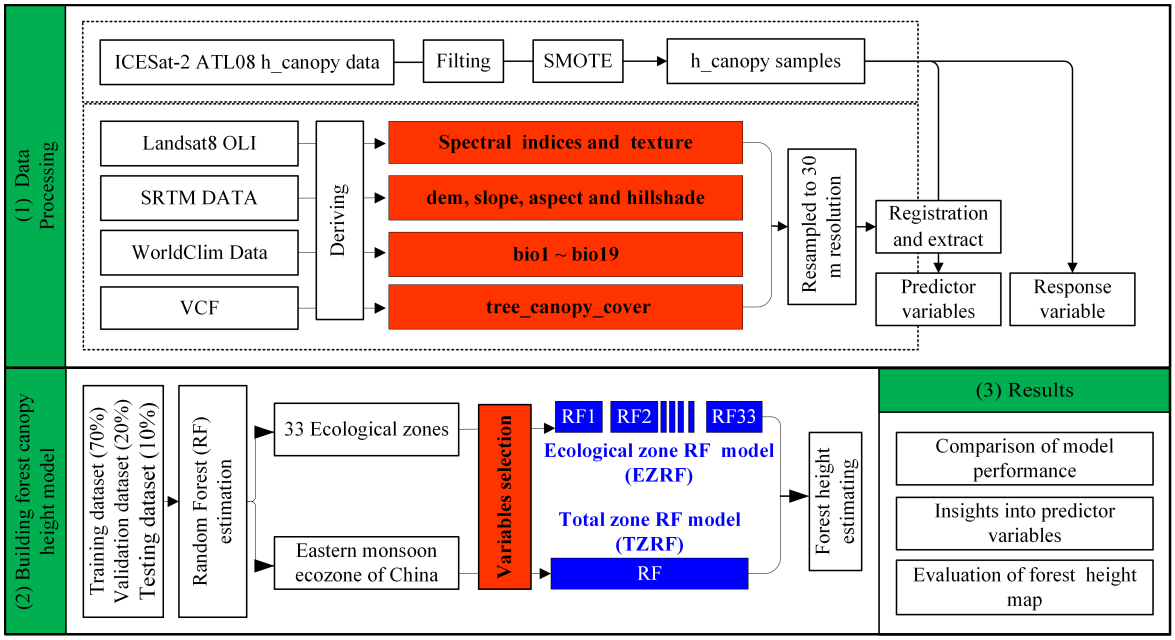


Fig. 2. Graphical summary of the main methods and steps performed in the study.

azimuth (in radians) of the sun; 3) the zenith (in radians) of the sun; 4) the slope calculated from terrain height; and 5) the aspect data from terrain height [46].

3) Landsat Tree Cover Percentage Product: Considering the forest coverage on forest growth, we adopted the Landsat vegetation continuous field (VCF) product as a variable for FCH modeling. The VCF represents the percentage of woody plants that cover the horizontal ground, with a spatial resolution of 30 m [47]. In this study, we used the Landsat VCF 30-m product provided by GEE. Its data ID is "NASA/MEASURES/GFCC/TC/v3," and the 2015 data were extracted. Table III lists the details of all the ancillary data.

III. Methods

Considering that forest height models with large coverage areas may not be applicable to all sites in all biomes, we proposed an EZRF approach that allows us to combine multiple FCH model into one final composite solution for multiecological zone forest height prediction. Fig. 2 provided the graphical summary of the main methods and steps we implemented in this study. First, we filtered ATL08 h_canopy data as the predictor variable and collected a series of variables (Landsat variables and ancillary variables) corresponding to forest height growth as predictive variables. Second, we developed FCH models for each ecological zone as the EZRF model and for the total eastern monsoon ecozone as the TZRF model. After the model prediction, ATL08 h_canopy data, airborne LiDAR data, and the Global FCH, 2019 [20] were used to assess the model's performance in predicting FCH. Finally, we compared the performance of the EZRF and TZRF models, provided insight into the impact of predictor variables on FCH estimation, and evaluated the accuracy of the predicted forest height map.

TABLE III
Ancillary Data

Type	Variable name	Description
Climate	bio1	Annual Mean Temperature
	bio2	Mean Diurnal Range
	bio3	Isothermality
	bio4	Temperature Seasonality
	bio5	Max Temperature of Warmest Month
	bio6	Min Temperature of Coldest Month
	bio7	Temperature Annual Range
	bio8	Mean Temperature of Wettest Quarter
	bio9	Mean Temperature of Driest Quarter
	bio10	Mean Temperature of Warmest Quarter
	bio11	Mean Temperature of Coldest Quarter
	bio12	Annual Precipitation
	bio13	Precipitation of Wettest Month
	bio14	Precipitation of Driest Month
	bio15	Precipitation Seasonality
	bio16	Precipitation of Wettest Quarter
	bio17	Precipitation of Driest Quarter
	bio18	Precipitation of Warmest Quarter
	bio19	Precipitation of Coldest Quarter
Topography	DEM	Elevation above sea level in meters
	Slope	Slope in radians
	Aspect	Aspect in radians
	Hillshade	hd1, hd2, hd3, hd4, hd5, hd6
VCF	tree_canopy_cover	The percentage of pixel area covered by trees

A. Ecological Zoning Modeling

The FCH in different terrestrial ecosystems varies significantly from region to region, influenced by prevailing climate and topographic conditions, and there may be a complex

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nonlinear relationship between the FCH and remote sensing parameters. A single model cannot accurately estimate the tree height on a large scale. Traditionally, unphysical grid methods have been used in FCH modeling, such as regular-size grids [20], which did not consider the distinct nature of forest growth in different ecological zones. To model the complex relationship between remote sensing variables and forest height in a large scales of China, we utilized the ecological zone division system by Fu et al. [48], and the eastern monsoon ecozone was divided into 33 ecological zones, as shown in Fig. 1(b), including deciduous conifer forest (01); mountainous conifer and broad leaf forest (02 and 04); agriculture (03, 05, 09, 11, 13, 16, 17, and 29); meadow and grassland (06); coniferous BLF and forest grassland (07); hilly deciduous BLF (08); mountainous deciduous BLF (10); agriculture and grassland (12); hilly evergreen BLF (14); mountainous deciduous and evergreen BLF (15); the Three Gorges Reservoir (18); agriculture and MF (19); mountainous evergreen BLF (20, 23, and 25); mountainous and hilly evergreen BLF (21, 22, 26, and 28); evergreen BLF (27); tropical monsoon forest and rainforest (30 and 31); tropical agriculture (32); and mountainous tropical rainforest and monsoon forest (33). Ecological zones represent natural conditions, such as different landforms, hydrothermal combinations, and vegetation characteristics, and building ecological zone-based models is one of the approaches to improve the accuracy of FCH estimation. We tested the ecological zone-based model approach for FCH estimation in 33 ecological zones in China, to reveal the peculiar relationships in forest growth across different ecological zones.

B. ZRF and TZRF Model Building

The EZRF model and the TZRF model were built based on the RF algorithm. RF was used in this study due to its nonlinear characteristics and strong adaptability to various datasets and situations [49]. We used the partitioned datasets and the complete datasets, respectively, as the predictor variables for the following: 1) the RF model was established independently in each ecological zones as the EZRF model and 2) the RF model was established in a total eastern monsoon ecozone as the TZRF model. The response variables for the EZRF model and TZRF model were the ICESat-2 ATL08 h_canopy data, and the corresponding predictor variables were the Landsat variables and ancillary variables. In this study, a total of 62 predictor variables related to FCH estimation were obtained; to evaluate the contribution of different categorical variables to the model performance in different ecological zones, we compared models using only Landsat variables (33 variables), using only ancillary variables (29 variables), and using all variables (63 variables). Since most of the ATL08 h_canopy sample is concentrated at 10–20 m, there are very few samples below 10 m and few above 20 m. Imbalanced data will cause models to ignore information from minority samples and have poor prediction performance. Here, we used the synthetic minority oversampling technique (SMOTE) [50] to analyze and simulate minority h_canopy samples. The ATL08 h_canopy samples in each ecological zone were classified at 2, 4, and 6 m intervals, and SMOTE oversampling and model training are performed for each classification scheme.

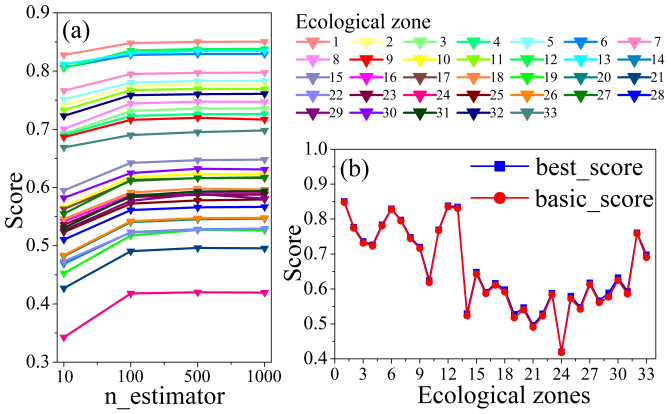


Fig. 3. Results of the parameter selection for the n_estimator. (a) Model scores of each ecological zone when n_estimator in 10, 100, 500, and 1000. (b) Comparison results of the best_score (selected n_estimator) and the basic_score (n_estimator = 100) of each ecological zone.

Parameter selection is needed in the RF models. In this study, we only used one key parameter of the RF model, namely, n_estimator, to correspond to the numberOfTrees parameter in the GEE [51]. We trained the RF model offline (in Python sklearn) using the selected training dataset, and the grid search method (cv = 5) was used to obtain the models score of each ecological zone when n_estimator in 10, 100, 500, and 1000 [as shown in Fig. 3(a)]. The highest score of different n_estimators for each ecological zone was selected as the best_score, and the score when n_estimator equals 100 (default parameter value) as the base_score; according to the comparison results of the best_score and the basic_score [Fig. 3(b)], there were small difference between the best_score and the basic_score, and we set the n_estimator parameters as default in the later EZRF and TZRF modeling process.

C. Variables Selection

Variable selection is an important process in improving model efficiency, which can make the model more interpretable. As mentioned earlier, in order to retrieve an accurate FCH map, we selected a total of 62 variables related to forest height. To reduce the model complexity and obtain the optimal variable domain, we chose a different number of variables in different ecological zones of the EZRF and TZRF models, to obtain the highest model accuracy and model efficiency. To be explicit, we added variables in the EZRF and TZRF models in order of variables importance ranking, first adding the variable with the highest importance score and calculating the r-value using the test dataset, and then adding the first two most important variables, the first three most important variables, …, cycle through the process. The contribution of each variable in the EZRF and TZRF models was measured by the percent increase in mean square error (%IncMSE). Nevertheless, the value of the variable contribution differed in the EZRF and TZRF models training [52]. To obtain a stable variable importance ranking, we trained the model for 25 epochs in each ecological zone and finally

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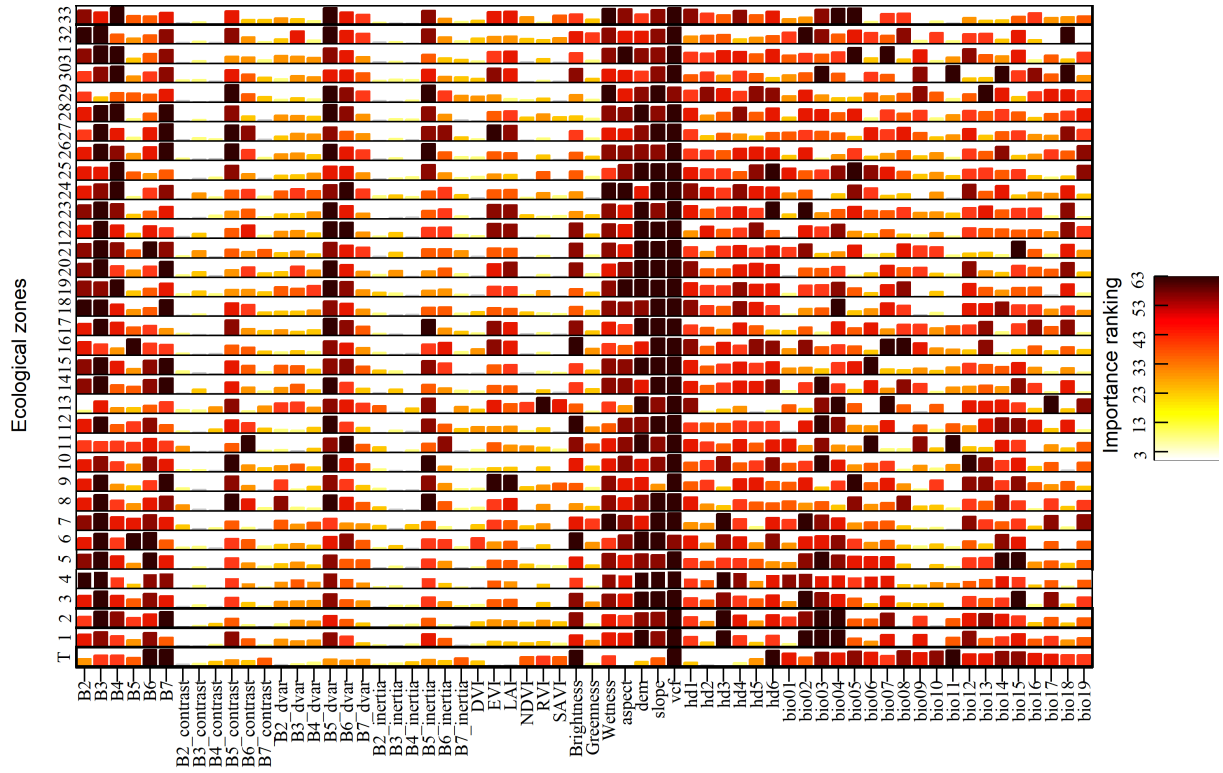


Fig. 4. Variable importance rankings identified in the TZRF model (ecological zone: T) and the EZRF model (ecological zones: 1–33). The warmer the color, the more significant the variable.

obtained the variables ranking in each ecological zone. The variables importance ranking identified in the TZRF model and the ZERF model were shown in Fig. 4. Since the variable selection cannot be executed on the GEE platform, according to the offline training and online prediction (OTOP) framework proposed by Wang et al. [53], we trained and tested the EZRF and TZRF model in GEE using selected variables obtained from offline training and generated a continuous FCH map.

D. Accuracy Validation and Comparison

Our predicted FCH was validated using the set-aside ATL08 h_canopy data and the airborne LiDAR forest height data. The accuracy of the predicted height was assessed based on the root mean square error (RMSE), which is defined in (2) and was used to quantify the differences between the paired datasets. The correlation coefficient (r) provides information on the degree of similarity of two datasets and can be defined as in (3) [54]. The mean bias estimate (MBE) was computed as in (4) to provide a measure of under- or overestimation. The RMSE, r, and MBE were calculated as follows:

$$\mathrm{RMSE} = \sqrt{\sum_{i=1}^{n}(h_{ai} - h_{mi})^2/n} \tag{2}$$

$$\mathrm{r} = \sum_{i=1}^{n}\left(h_{a_i} - \overline{h_a}\right) \times \left(h_{m_i} - \overline{h_m}\right) \Big/ \sqrt{\sum_{i=1}^{n}\left(h_{a_i} - \overline{h_a}\right)^2 \sum_{i=1}^{n}\left(h_{m_i} - \overline{h_m}\right)^2} \tag{3}$$

$$\mathrm{MBE} = \sum_{i=1}^{n}(h_{ai} - h_{mi})/n \tag{4}$$

where n is the number of validation data pairs; a represents the airborne LiDAR data, and m represents the model prediction data. h_{ai} is the ATL08 h_canopy data or the airborne LiDAR forest height data. h_{mi} is the predicted FCH, and $\overline{h_a}$ and $\overline{h_m}$ are the average values of h_{ai} and h_{mi}, respectively.

IV. Results

A. Variables Importance in Different Ecological Zones

Fig. 4 shows the variables importance ranking identified in the TZRF model (ecological zone: T) and the EZRF model (ecological zones: 1–33). A total of 33 variables were obtained from Landsat images and 29 from ancillary data. We can see that the variables showed a similar importance ranking in different ecological zones, and that the importance of individual variables will vary depending on the geographical location of the ecological zones, as indicated by the color distribution in Fig. 4. Tree canopy cover was considered as an important variable in both the TZRF and EZRF models, with an average feature score of 66.32 across the 33 ecological zones (Fig. 5), but its effect was not consistent across all ecological zones. Such as in ecological zones 06 and 29, which was domained by agriculture and with a sparse forest coverage, the importance of tree canopy cover was slightly diminished. According to the GLC_FCS30-2020 [29], forest communities in different ecological zones are structurally diverse, ranging from closed forests, open forests, and sparse forests. For each

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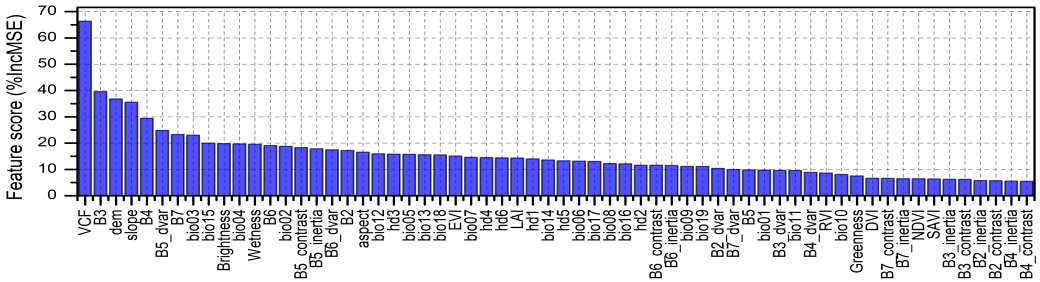


Fig. 5. Variable importance scores for the EZRF model, averaged over 33 ecological zones.

of the 33 ecological zones, a clear relationship between FCH and forest cover was evident.

Terrain variables, including aspect, dem, and slope, were the second most important ancillary variables, with the average scores of 35.51–36.74 (Fig. 5). In addition, the hillshade variables, which were used in FCH estimation for the first time, have shown certain importance. It can be seen obviously from the EZRF model (Fig. 4), particularly, that hd3 and hd4 (the hillshade variables when the azimuths were 120° and 180°) were more important than the Landsat texture variables (B2 texture, B3 texture, B4 texture, and B7 texture) and spectral index variables (RVI, DVI, SAVI, NDVI, and TCG), with an average score of 19.4 (Fig. 5). The TZRF model showed an opposite result that none of the hillshade variables were particularly important. We thought this maybe because the large-scale models failed to reflect the forest growth difference with the sunlight shift in China's south and north, whereas the EZRF model was able to reflect the effects of different ecological zones on forest growth, including the influence of hillshade (caused by changes in sunlight) on forest.

The importance of climate variables in the EZRF model varied significantly with ecological zones (Fig. 4). On the whole, the importance of temperature variables (bio1–bio11) varied with ecological zones, while the influence of precipitation variables (bio12–bio19) on FCH was relatively uniform in different ecological zones. Overall, bio4 and bio15 were the most important temperature and precipitation variables for FCH estimation, with the mean variable scores of 19.67 and 22.99 (Fig. 5), respectively. The result showed that water was the driving force for tree growth on large scale, while temperature effects were local differences. Tree growth is affected by both of these factors and exhibits a large number of factors that affect the different distribution of trees in different ecological zones.

Shading may affect the spectral reflectivity in Landsat pixels of forest communities in different ecological zones. However, Landsat spectral variables were also important in the FCH estimation. Obviously, the Landsat raw band (B2–B7) was the important Landsat variables in both the EZRF and TZRF models (Fig. 4); among them, the green band (B3) was the most important for the EZRF model, with a variable score of 39.52 (Fig. 5). The shortwave infrared bands (B6 and B7) were the most important Landsat variables for the TZRF model. However, in the EZRF model, the B6 variable showed a better performance in the northern ecological zone (1–14), while the importance decreased in the southern ecological zone (15–33). This may be due to the fact that the reflectivity of the SWIR increases with decreasing leaf water content. The near infrared (NIR) (B5) texture variables B5_Contrast, B5_Dvar, and B5_Inertia were better performance than the raw band B5 in different ecological zones, while the spectral indices (DVI, NDVI, RVI, and SAVI) showed a minor importance in the different ecological zones, reflecting the importance of spatial variables were more important in the FCH estimation than spectral variables, which represent the horizontal information of forest. The TC transformation, especially the brightness and wetness, was important in FCH estimation, which were crucial indicator of canopy maturity and structure and appear sensitive to the interplay between water content and canopy structure [40].

B. Comparsion ZERF and TZRF Model Performance

1) ATL08 H_Canopy Data Resampling: Before modeling, we analyzed the samples and found significant imbalances between forest height categories. Sample sizes with the tree heights below 10 m and above 20 m were in the minority in the sample set, resulting in the models that tend to be biased in their predictions. So, we explored using different height intervals for the SMOTE algorithm, as shown in Fig. 6. Regarding the effect of the sampling interval on model accuracy, oversampling at different intervals (6, 4, and 2 m) had little effect on the training results of the model. Fig. 6 shows that out of the three sampling intervals, 6 m yielded the highest r in ecological zones 1–9 and 4 m yielded the highest r in ecological zones 10–33. These results indicate that the 4- and 6-m intervals perform well in the southern and northern ecological zones, respectively. To reduce the computational cost, we preferred 5 m as our final ATL08 h_canopy data sampling interval.

2) Variable Grouping Modeling: In addition to the variable importance measures, we evaluated the contribution of different groups variables to FCH modeling, by comparing models using only Landsat variables (33 variables), using only ancillary variables (29 variables), and using all variables (63 variables). The orange squares indicate Landsat variables, and we can see that the Landsat variables alone performed better than the green triangle (ancillary variables) in ecological zones 10–33. The EZRF model using only Landsat variables resulted in a comparatively higher r (average $r = 0.65$) than that using ancillary variables alone (average $r = 0.61$, Fig. 6).

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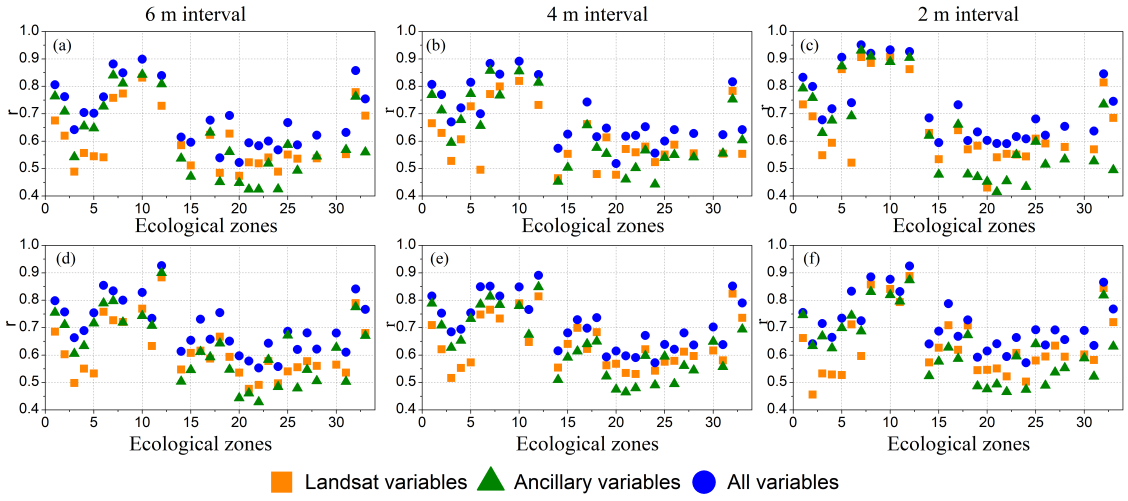


Fig. 6. Correlation coefficient (r) of the EZRF model on validation datasets. (a)–(c) NLF. (d)–(f) BLF. The training data for each ecological zone were resampled by SMOTE at the intervals of 6, 4, and 2 m, respectively.

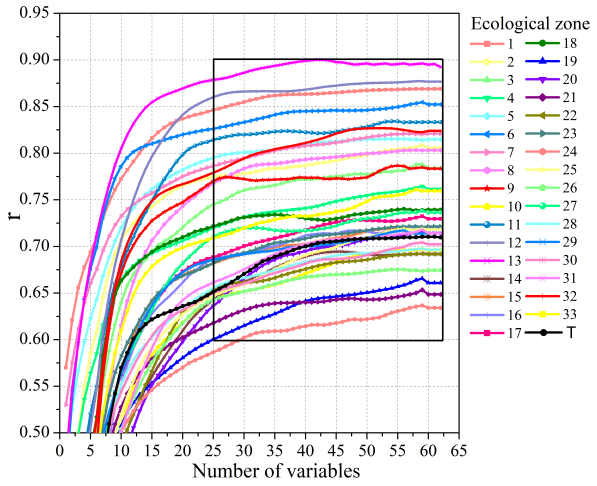


Fig. 7. Accuracy of the TZRF model (ecological zone: T) and the EZRF model (ecological zones: 1–33) varied by the number of predictor variables.

Regarding the influence of the combination of predictors on model accuracy, the combined datasets (Landsat + ancillary variables) yielded the best performance in all ecological zones [Fig. 6 (blue dots)], with the highest r (average $r = 0.72$). In addition, the EZRF model using only the Landsat variables had the lowest r in the northern ecological zones (Fig. 6; orange box of ecological zones 1–9). Considering that multivariables participation in modeling is one way to improve the FCH estimation accuracy, we selected the combined variables (Landsat + ancillary variables) in the next modeling step.

3) Variable Selection of EZRF and TZRF Models: For the FCH estimation involving the use of joint Landsat and ancillary variables, we performed variable selection to balance model accuracy and practical operational efficiency. According to the variable importance ranking in Fig. 4, variables were put into the model in sequential increments, and moreover, r was calculated for each model with a different number of variables. The results were shown in Fig. 7, where the accuracies of the TZRF model and the EZRF model were varied by the number of predictor variables. We can see that when the number of predictor variables is less than 25, the prediction accuracy r increases considerably with the number of variables. When the number of predictor variables is larger than 25, the rate of improvement in prediction accuracy slows down, and the model accuracy r becomes more stable, as the number of variables increases, as shown in Fig. 7 (black box). Ideally, the use of more variables would better represent the variable domain and, thus, improve the model accuracy. To obtain the number of variables corresponding to the largest r for each model, we zoomed each curve in Fig. 7 (black box) to within 0.01 accuracy of r, as shown in Fig. 8. Clearly, the model accuracy increased with the number of variables, at least to some extent. We can see that the TZRF model achieved an optimal r of 0.71, when the number of predictor variables reached 47. In the EZRF model, the number of predictor variables needed to achieve optimal modeling accuracy was different for each ecological zone. Overall, the EZRF model achieved better accuracy in terms of r than the TZRF model in the first 18 ecological zones. While, the model accuracies of EZRF in 19–31 ecological zones were lower than that of the TZRF model (as shown in Table IV). It can also be observed from Table IV that the EZRF model is more likely to achieve superior prediction accuracy in average (average r of the EZRF model is 0.80) than the TZRF model, despite the lower accuracy of the individual ecological zones.

4) Evaluation of ZERF and TZRF Model by ATL08 Data: Fig. 9 shows scatter plots of the ATL08 h_canopy data versus our predicted FCH from the EZRF and TZRF models, respectively. For the EZRF model, the results showed a lower MBE (−0.04 to 0.06 m) in different forest types. We achieved an RMSE of 2.51 m and an r of 0.93 for MF [Fig. 9(c)]. We found that there was usually greater uncertainty in areas with very low or very high FCHs in the BLF and NLF. The RMSEs of the BLF and NLF were 3.77 and 3.13 m, respectively, and the r values were 0.86 and 0.88, respectively. When the FCHs were less than 10 m, the predicted heights were overestimated

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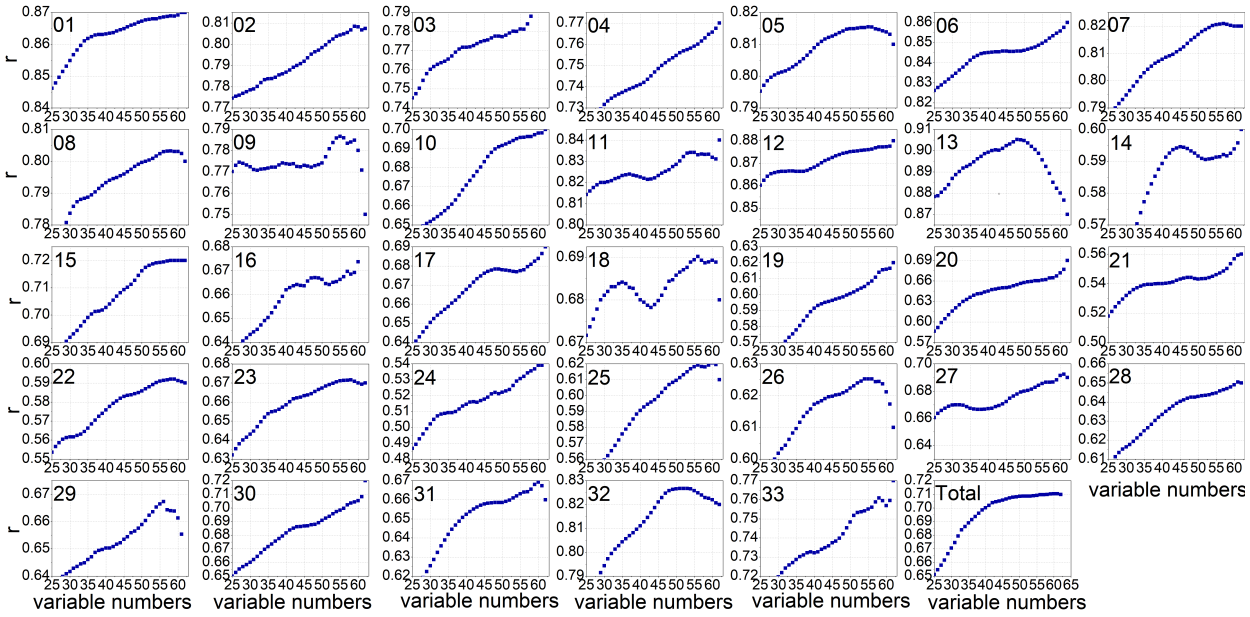


Fig. 8. Local magnification of Fig. 7 (black box part), to observe the variation tendency of model accuracy r with the number of variables on the 0.01 scale.

TABLE IV
SELECTED VARIABLE NUMBERS OF THE TZRF MODEL (ECOLOGICAL ZONE: T) AND THE EZRF MODEL (ECOLOGICAL ZONES: 1–33)

Ecological zone	number of variables	r	Ecological zone	number of variables	r
1	46	0.87	18	50	0.74
2	57	0.81	19	54	0.66
3	45	0.78	20	49	0.71
4	51	0.76	21	55	0.65
5	37	0.81	22	50	0.69
6	40	0.85	23	48	0.72
7	49	0.82	24	53	0.63
8	44	0.80	25	54	0.72
9	27	0.78	26	39	0.67
10	54	0.70	27	54	0.74
11	47	0.83	28	56	0.70
12	50	0.88	29	47	0.71
13	37	0.90	30	53	0.70
14	38	0.69	31	42	0.71
15	50	0.72	32	43	0.82
16	46	0.72	33	55	0.76
17	45	0.73	T	47	0.71

(average MBE = 1.73), and when the FCHs were greater than 20 m, the predicted heights were underestimated (average MBE = −2.78). For the TZRF model, the scatter plots for the corresponding forest types displayed comparatively discrete distributions [Fig. 9(d)–(f)], possibly due to the poor performance of individual ecological zones, and the TZRF model failed to capture these individual errors. The MLF had the highest r value of 0.8, which was lower than the accuracy of the EZRF model, and for the BLF and NLF, the accuracy was lower, with r ranging from 0.75 to 0.77. In particular, the predicted FCH of the TZRF model was significantly overestimated at forest heights below 10 m, with an average MB of 3.32 m, while the overestimation of the EZRF model was relatively minor. In terms of all forest types, the TZRF model clearly underestimates the forest height at heights greater than 20 m, with an average MB of 4.62 m, and the EZRF model has lower errors than the TZRF model, with RMSE differences ranging from 1.03 to 1.44, and the results shown that the EZRF model is 21.5%–36.5% more accurate than the TZRF model. The results indicated that our proposed EZRF model outperforms the traditional TZRF model, which can quantify the error in a single ecological zone; thus, higher prediction accuracy and better variable interpretability can be achieved.

C. FCH Mapping and Evaluation With the EZRF Model

1) FCH Mapping by the EZRF Model: FCH was predicted across the 33 ecological zones using the EZRF model. The resulting 30-m resolution FCH map of the eastern monsoon ecozone in China was shown in Fig. 10; according to the forest region defined by the GLC_FCS30 [29], areas of nonforest or with FCH values less than 3 m were masked [Fig. 10 (white region)]. The results indicate that the FCH for all the eastern monsoon ecozones was in the range 2.41–34.98 m. The insert map in Fig. 10 shows a zoomed-in view of the detail. Fig. 10(a)–(c) was selected from three different ecological zones, and Fig. 10(a) was located in a deciduous coniferous forest ecological zone in the northern mountainous area of Greater Khingan Mountains; due to climatic factors, the forest heights below 14 m and above 20 m are discrete and clearly distributed in sheets. Fig. 10(b) was located in a deciduous and EBLF ecological zone in Qinba Mountain, located at the north–south boundary of the Chinese climate zone. The area has abundant forest resources and a high vertical vegetation distribution; Fig. 10(c) was located

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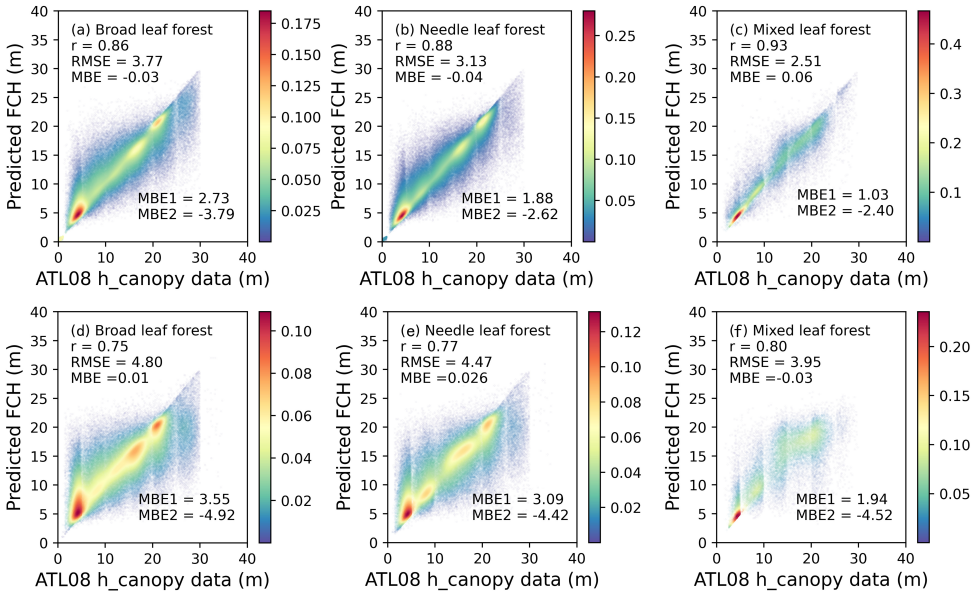


Fig. 9. Scatter plots of ATL08 h_canopy samples versus our predicted FCH by (a)–(c) EZRF model and (d)–(f) TZRF model. MBE1 is the MBE value when the FCHs were less than 10 m, and MB2 is the MBE value when the FCHs were greater than 20.

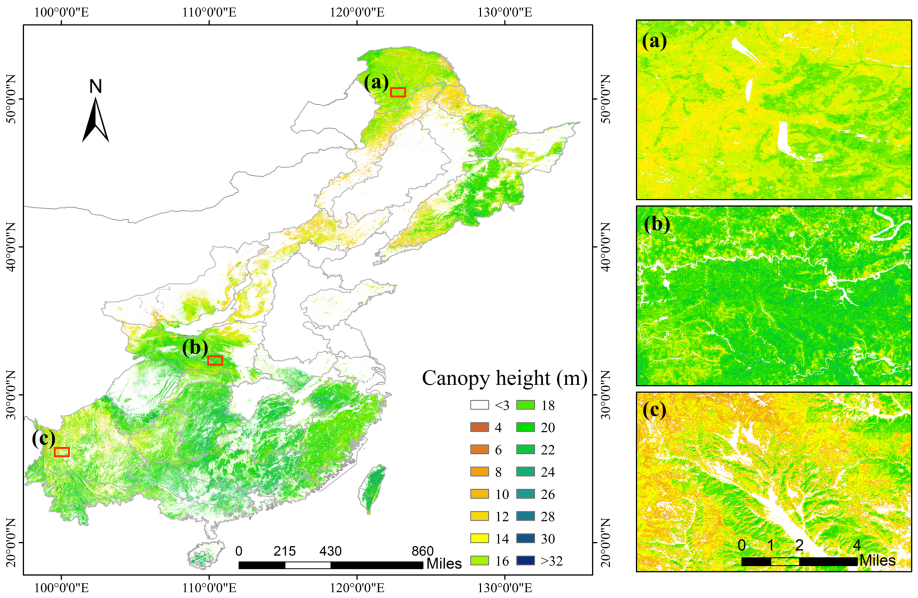


Fig. 10. Distribution of FCHs in the eastern monsoon ecozone of China. (a)–(c) represent the zoomed-in view part of the FCH map from three different ecological zones, which displayed detailed information of FCH distribution.

in the western half of the Yunnan–Guizhou Plateau, covering the southern part of the Hengduan Mountains and the mountainous area of southwest Sichuan. The height of the forest ranged from 3.30 to 27.39 m, and the distribution along the gradient of elevation can be clearly observed, with the height of the forest being higher along the Nujiang River in the southwest and lower in the valleys due to the presence of provincial roads and towns. Overall, our estimated FCH was strongly influenced by the geographical distribution of forests, forest types, and forest height distribution, with high spatial heterogeneity evident. The FCH was relatively high in the southern tropical rainforest, monsoon forest, northeastern Changbai Mountain, Qianshan Mountain NLF and BLF zones, and low in the North China Plain and northwestern semiarid regions, which was high consistency with the spatial pattern of FCH reported by Huang et al. [26].

2) Map Validation by Airborne LiDAR Data: In 2021, we collected airborne LiDAR data from five sites in the southern ecological zone of China, with tree heights ranging from 1.3 to 28.7 m, with an average value of 7.69 m. The airborne LiDAR validation results are shown in Fig. 11. Overall, the predicted FCHs were somewhat overestimated, with MBEs ranging from −0.92 to −4.29 m across the five sites. The five panels in Fig. 11 represent the scatter

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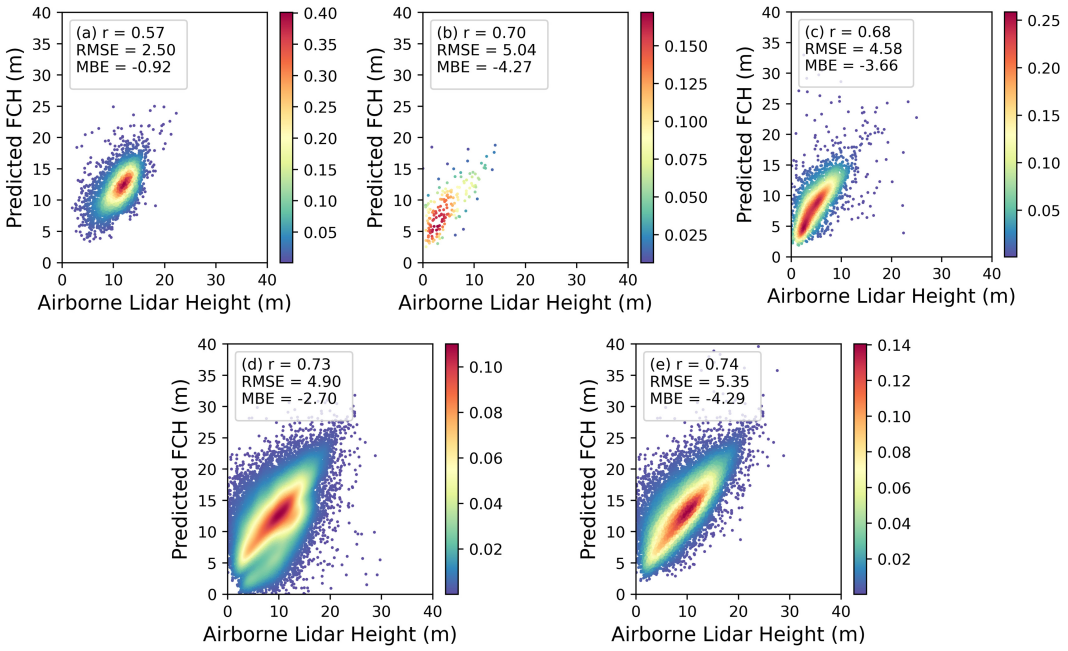


Fig. 11. Scatter plots of the airborne LiDAR height versus our predicted FCH. (a)–(e) represent the five airborne lidar sites as shown in Table 1.

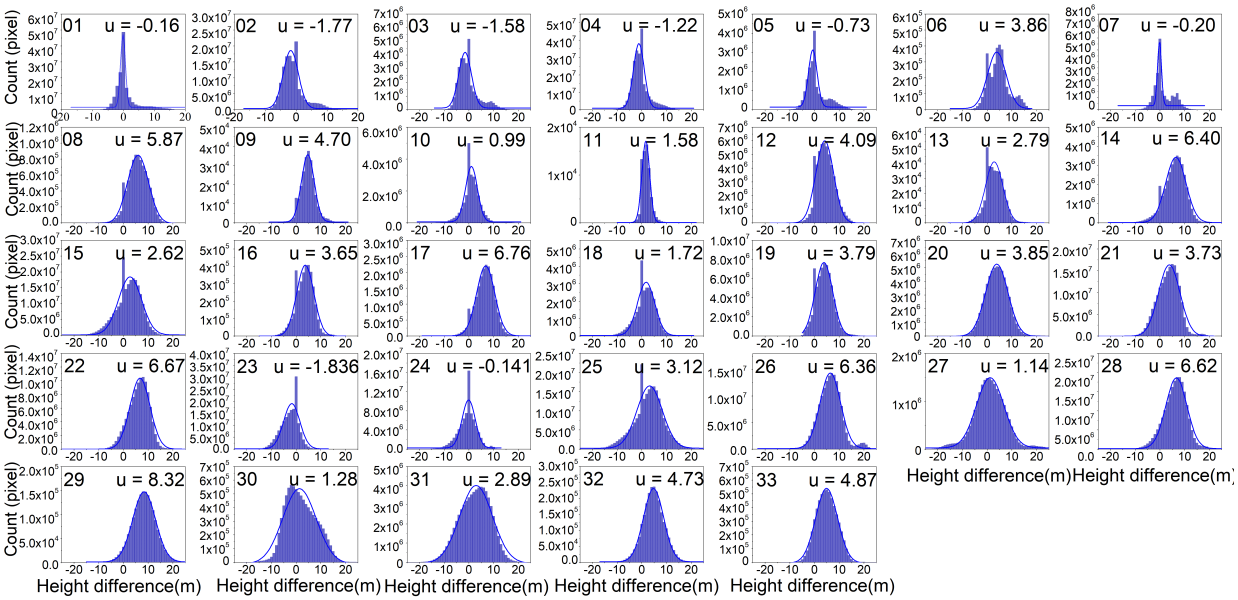


Fig. 12. Frequency distribution across the 33 ecological zones of the height difference (our predicted FCH minus the global forest height product of Potapov). The variable u represents the mean forest height difference, which was obtained from a Gaussian fitting curve.

plots of the airborne LiDAR height versus our predicted FCH; Fig. 11(a)–(e) corresponds to the five regions in Fig. 1. For the karst evergreen BLF ecological region [Fig. 11(a)], the RMSE was 2.50 m, and r was 0.57. The areas in Fig. 11(d) and (e) are located in mountainous evergreen BLF ecological zones. The RMSE in these areas (4.90 and 5.35 m) was more sensitive than that in other areas due to forest height deviation [Fig. 11(d) and (e)], and the r values, 0.73 and 0.74, revealed the consistency of the predicted FCH with the airborne LiDAR data used for validation in these two areas.

3) Comparison FCH Map With the Global FCH: Since there were no forest height products with the same resolution in China for the most recent year, the global forest height product of 30-m resolution in 2019, produced by Potapov et al. [20], was used for comparison with our FCH results. Fig. 12 shows the distribution of forest height differences between these two height products. We discovered that our estimated FCH was generally, but not universally, higher than the global forest height product of Potapov et al. [20] in the 33 ecological zones. This may be because we used the 98% height from ICESAT-2 ATL08, rather than the 95% height from GEDI. The differences in forest height were not uniform across all ecological zones. The variable u represents the mean forest height difference for each ecological zone and was obtained from a Gaussian fit curve. As shown in Fig. 12, the mean height difference varied from −1.77 to 8.32 m, with an average value of 2.87 m across all ecological zones. Ecological zones 1–5 showed an underestimation phenom-

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enon, with u values below 0. These ecological zones are located in northeastern China, and the forest types are mostly NLF and deciduous BLF. Ecological regions 8–33 showed an overestimation phenomenon, except for individual zones (23 and 24); these ecological zones are mostly located in a montane BLF area, with dense forest, and the year-round weather is more rainy, which will affect the LiDAR laser penetrate the ground.

V. Discussion

A. Insights Into Predictor Variables

1) Variable Selection: In this section, we focused on the driving factors in the modeling of FCH and the internal mechanisms that predict the variables in different ecological zones. To improve the predictive efficiency and interpretability of the RF model, we perform variable selection based on the importance ranking of the 62 variables, in a sequential progressive manner, and select the number of variables corresponding to the optimal model accuracy for each ecological zone. Since we used the GEE cloud platform, the actual modeling time was significantly reduced. However, from the curve of the model accuracy varied with the number of variables, when the number of the variables reached half (30 or so), the model accuracy is almost essentially satisfied with the requirements, which will greatly shorten the modeling time and, thus, improve the efficiency of forest height mapping. Therefore, in practical applications, we need to fully account for both accuracy and efficiency in order to better serve practical production studies.

2) Ancillary Variables Analysis: Among all the selection variables, canopy coverage shows the most significant performance in both the EZRF and TZRF models. According to Nandy et al. [55], tree canopy cover impacts FCH by directly affecting the photon penetration rate of ATLAS. This greatly enhances the discrimination of spatial patterns of forest cover, especially in highly fragmented landscapes and agricultural regions. This observation supports the concept that canopy cover explains much of the forest height modeling, which is in accordance with the view of Lefsky [19]. Dem and slope were also considered meaningful in all ecological zone models, possibly due to the high correlation between elevation and temperature seasonality (bio4). In addition to this factor, steeper topography enhances tree biomechanical damage via gravitational forces and influences wind disturbance that could limit tree height [56]. Furthermore, topography is also key in soil development and erosion, which, in turn, affects soil water retention, playing a key role in the patterns of forest structure [57]. According to Wang et al. [22], mean annual precipitation, soil moisture, and many other characteristics important for plant growth were significantly more favorable at lower elevations and lower slopes than at higher elevations and slopes.

Moreover, in conjunction with field surveys, we have found that trees always exhibit a different height distribution on the two sides of a mountain. In our results, we found that the hillshades of solar azimuths in the south (hd4, which means 180°) and southeast (hd3, which means 120°) were more conducive to forest growth than those in other directions, especially in the northern ecological zones (Fig. 5). Forest growth is strongly correlated with the amount of radiation received on the Earth's surface. Solar radiation is associated with local obscuration. In the northern ecological regions, slopes facing the north receive less solar radiation than slopes facing the south, so forests grow in poorer conditions [53], [55]. However, the phenomenon was weaker in the southern ecological regions, possibly due to the stronger water transpiration due to the strong solar radiation on the sunny side of the mountain, and the lack of water affecting tree growth. Overall, the ancillary variables perform better than the Landsat variables in modeling the forest height, and the addition of the Landsat variables improves the accuracy of the RF model in FCH estimation, as shown in Fig. 6.

3) Landsat Variables Analysis: Green reflectance was the most important Landsat variable for FCH modeling and was also found by Donoghue and Watt [12]. The green spectral band has a biophysical basis, which is related to chlorophyll II [3]. In the southern ecological zones (26, 27, and 28), where forest canopy was dense, leaf water content may contribute to the importance of SWIR1 and SWIR2. Other studies have shown that SWIR is helpful in predicting FCH, as it is less affected by atmospheric humidity than other wavelengths. The NIR texture provides rich information about the forest and can improve the estimation of the forest height. However, the original spectral band NIR was of low importance due to shadowing effects caused by the complexity of the forest canopy and emergent trees in the ecological region [58]. The RVI, NDVI, and SAVI are usually used to describe the status of vegetation growth [40]. However, they encounter some problems when used for FCH modeling. The saturation of vegetation indices often occurs in dense forest areas, which directly leads to model insensitivity to forest structure [7]. In addition, some vegetation indices are sensitive to the shadow phenomenon of complex or mountainous terrain due to the different combinations of spectral bands, which indirectly leads to model insensitivity to vegetation structure [59].

The importance of the predictive variables was different in different geographical environments, and the effects of ancillary variables in FCH modeling alone were better than those Landsat variables (Fig. 6). This is mainly due to the fact that Landsat data alone do not provide good predictors for FCH modeling due to the low penetration of Landsat bands, and their combination with ancillary data yields better results than remote sensing data alone. This may be indicative of the mixed variables in the interaction potential of the prediction process, as well as the ability of RF algorithms to learn the relationship between forest height and predictor variables from the complex interactions between remote sensing variables and ancillary variables.

B. Insights Into Ecological Zoning Methods in FCH Mapping

1) Ecological Zoning Method Analysis: In the following, we highlight the challenges and opportunities presented by the new generation of ICESat-2 ATLAS in mapping the FCH over large areas of diverse ecosystems. Differences in ecosystems require RF models to capture local forest conditions, so we construct RF models for each ecological zone

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as the EZRF model and the RF model on the total zone as the TZRF model, in order to explore which model methods are suitable for large-scale forest height estimation. From the analysis of the results in Table IV, we found that the approach based on ecological zoning modeling was able to quantify the error in forest height within a single ecological zone and, thus, define the uncertainty of the model under different ecological conditions. From the perspective of model evaluation, the accuracy of forest height estimation is better for the ZERF model than for the TERF model. The TZRF model is prone to wrap uncertainties from individual ecological zones, which reduces the overall accuracy of the model performance. According to Lefsky [19], there are potential advantages to utilizing ecological zoning-based methodology. The ecological zoning method can be used for spatial segmentation based on ecosystem types and geographical features related to forest growth to make the analysis of FCH features practically significant and to make the characterization of forest heights more practical instead of applying arbitrary geometrically scaled zoning. Ecozone-based modeling approaches are similar to ensemble models in that they are highly accurate, fast to train, insensitive to outliers, and suitable for large-scale forest height estimation.

2) Evaluation of Ecological Zoning FCH Map: Our study area consists of 33 ecological zones extending from north to south across the eastern monsoon ecological zone of China. Compared with those in previous studies of ICESat-2 FCHs [25], [55], the verification results with ATL08 h_canopy data showed that ICESat-2 can yield highly accurate FCH estimates under different forest types, especially in the relatively lower canopies of MF [Fig. 9(c)]. In BLF and NLF, the RF model overestimated height in shorter forests and underestimated height in taller forests [Fig. 9(a) and (b)]; this result is better than the FCH generated by Li et al. [25] (which yielded an r of 0.46–0.54). The performance was significantly better in the northern ecological regions (zones 01, 02, 03, 04, 05, and 07) than in the southern ecological regions of the eastern monsoon ecozone region of China, and the low prediction accuracy in the southern ecological zones was mainly due to the higher tree cover. Studies [31] have shown that the ICESat-2 ATLAS photons could not penetrate dense canopies, and multiple tree canopies usually prevent much of the laser energy from reaching the surface. In addition, the saturation phenomenon of remote sensing data results in potential estimation error of FCH in dense forests in the southern region [60]. According to the study of Huete [37], optical remote sensing data lose sensitivity when the forest density is high. Therefore, estimating the FCH in pixels with high vegetation coverage can be challenging. The number of ATLAS canopy height samples also played a decisive role in the modeling of forest heights. The southern subtropical monsoon wet climate region has year-round rain clouds. ATLAS photon intensity decreases in the presence of clouds, resulting in signal attenuation, which will result in the absence of photons within the ATL08 segment or a number insufficient to correctly model forest height [60], [62]. In contrast, this phenomenon was rare in the northern sparse canopy and NLF environments. In addition to the above factors, other factors, including atmosphere, the phenological state of vegetation, and terrain, impact accuracy in the forest height modeling process [23], [63].

3) Prospects of Ecological Zoning FCH Mapping: Furthermore, compared with the results obtained by the $1^\circ \times 1^\circ$ tile-based model [29], our results showed higher values in most ecological zones, and the RMSE (2.51–3.77 m) of our predicted forest height was 43%–62% more precise than that of the GEDI global FCH studies (RMSE was 6.6 m). This was expected; since the growth patterns of trees are related to ecological factors (geomorphology, geology, vegetation, and climate), it is necessary to describe forest dynamics by assuming ecological regional differences to improve the accuracy of prediction [64]. Our findings highlight the dominance of ecological zoning methods for FCH retrieval. Compared with global or continental models (Potapov's global model-based RMSE of 10.43 m), the limited geographically ecological zoning models establishing modeling coefficients associated with forest growth in each ecoregion may have higher accuracy. Although this work was based only on the ecological zone of the eastern monsoon zone in China, we are confident that this ecological zone-related modeling approach is applicable to other regions, such as studies in other ecological zones from the north to the tropics. The development of ecosystem-based regionalization models allows the capture and analysis of the unique relationships of forest growth of each ecological region in most forest conditions [65]. In addition, policy makers increasingly need ecological area studies and methodologies to help make strategic decisions and explore development options. Future efforts to extend this research to a regional, continental, or global scale will necessarily require the selection of remote sensing features and testing of models for different ecozones, as well as the rational segmentation and use of knowledge-based global ecozones [66], which will provide the best representation of locally homogeneous regions to build robust canopy height estimation models for global forest regions.

VI. Conclusion

This study showed that, the wall-to-wall FCH map (at 30-m resolution) using new-generation satellite LiDAR data (ICESat-2) is possible. Using the EZRF model, we explored the complex nonlinear relationship between forest height and predictor variables from Landsat variables and ancillary variables. The ecological zoning method was applied to 33 different ecological zones in the Eastern Monsoon Ecozone in China. The results showed that the following hold. First, the EZRF model was able to quantify the error in forest height within a single ecological zone and, thus, showed good performance than the TZRF model in forest height prediction at large scales, and the forest height map achieved the RMSEs of 2.5–5.35 m. Second, the importance of predictive variables in forest height modeling of different ecological zone was explored, and Landsat data alone perform good than ancillary alone, and the combination of Landsat variables with ancillary variables yields better results (with 13% improvement in accuracy) than Landsat variables alone. Third, we analyzed the importance of variables in different ecological zones and discovered first that the overall performance of the hillshade

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variable is better than that of the vegetation index in the FCH estimation, which reflects the effect of changes in sunlight and shadow on forest growth. We provided guidance on FCH modeling by analyzing the prediction accuracy of FCH in different sets of feature modeling, discussing the importance of features in different ecological zones, understanding the internal mechanisms of features in FCH estimation, and improving the predictive efficiency and interpretability of the models. Fourth, in addition, we explored the map errors, where the forest height map was more accurate in the northern ecological zone than in the southern ecological zone, mainly due to the weak ATLAS photon penetration and the dense forest coverage in the southern region. We strongly recommend that for the study of the ICESAT-2 forest height, the photons in regions with dense forest coverage should be considered to assess whether the photons reach the ground. The novelty of our work is to gain insight into the benefits of integrating ecological zoning in the FCH prediction. It is recommended to use a simple terrain height modification model (using other DEMs) to reduce the FCH error caused by inaccurate terrain height estimation. We anticipated that the proposed ecological zoning approach and variables insights will provide an opportunity to understand the carbon sinks of all large-scale forests.

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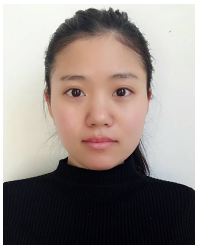
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Zhaocong Wu received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2004.

He is currently a Professor with the School of Remote Sensing and Information Engineering, Wuhan University, where he is also the Director of the Engineering Research Center of Communication, Teleportation and Integrated Application. He has authored or coauthored more than 100 scientific articles in refereed journals and proceedings, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS. His research interests include hyperspectral remote sensing and quantitative remote sensing, intelligent information processing of high-resolution remote sensing image, remote sensing satellite simulation and effectiveness evaluation, and application of remote sensing economics. His website is http://cnrs.whu.edu.cn/.



Fanglin Shi received the B.S. degree in remote sensing science and technology and the M.S. degree in photogrammetry and remote sensing from the School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo, China, in 2015 and 2018, respectively. She is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China.

She was a Visiting Student with the Institute of Atmospheric Physics, Chinese Academy of Science, Beijing, China, from 2016 to 2018. Her research interests include remote sensing image processing, forest parameter modeling from satellite light detection and ranging (LiDAR), and optical data.

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